Code No: 132AE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, March/April - 2023
ENGINEERING GRAPHICS
(Common to EEE, ECE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
- - -

1.a) Draw the hyperbola passing through the point P, assuming its asymptotes make an angle of 70^0 with each other. The point P is 40mm away from one asymptote and 45mm from the other.

b) A circle having a 50 mm diameter rolls within a circle with a 150 mm diameter with internal contact. Draw the locus of a point lying on the circumference of the rolling circle for its complete turn. Name the curve. [7+8]

OR

2.a) Draw the hyperbola when the focus and the vertex are 25 mm apart. Consider eccentricity as 3/2. Draw a tangent and normal to the curve at a point that is 35 mm from the focus.

b) Draw the locus of a point P moving so that the ratio of the distance from a fixed point F to its distance from a fixed straight line is 0.75. Point P is at a distance of 70mm from the straight line. Also, draw the tangent and normal to the curve. [7+8]

3. A straight line AB is 80 mm long. It is inclined at 45^0 to the HP and its top view is inclined at 60^0 to XY line. The end A of the line, which is farthest from VP is on the HP and 65 mm behind the VP. Draw the projections of AB and find its true inclination in relation to the VP if the line is in the third quadrant. [15]

OR

4. A circular plane of diameter 50 mm is resting on HP having a point on the circumference of the base with its surface inclined at 45^0 to VP and 30^0 to HP. Draw its projections. [15]

5. A pentagonal pyramid of 25 mm edge of base and 60 mm height is resting on the corner of its base on H.P. and the slant edge containing that corner is inclined at 45^0 with H.P. Draw the projections of the solid, when its axis makes an angle of 30^0 with V.P. [15]

OR

6. A cylinder of base diameter 50 mm and axis 70 mm has a point of its base circle in the V.P. Its axis is inclined at 30^0 to the V.P. and 45^0 to the H.P. Draw its projections. [15]

7. A pentagonal prism of base 30 mm side and 60 mm height is resting on the base in HP such that one of the rectangular face is parallel to the VP. It is cut by a plane perpendicular to VP and 60^0 inclined to HP and bisecting the axis of the solid. Draw development of lateral surface of the bottom part of the solid. [15]

OR

8. A square prism having a base with a 40 mm side and a 60 mm long axis rests on its base on H.P. such that one of the vertical faces makes an angle of 30^0 with V.P. A section plane perpendicular to the V.P. inclined at 45^0 to the H.P. and passing through the axis at a point 20 mm from its top end cuts the prism. Draw its front view, sectional top view. [15]

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9. Draw the orthographic projections of the part shown in figure 1.
a) Front view b) Top view c) Side view. All dimensions are in mm. [15]

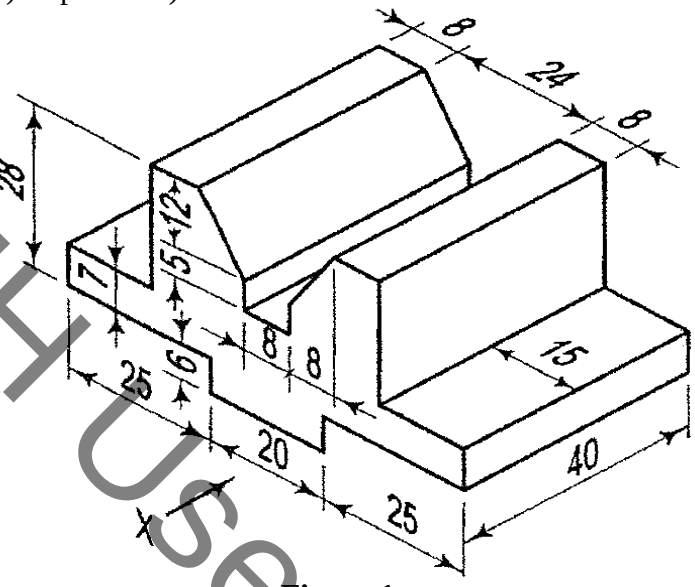


Figure 1

OR

10. Draw the isometric view of the following figure 2. All dimensions are in mm. [15]

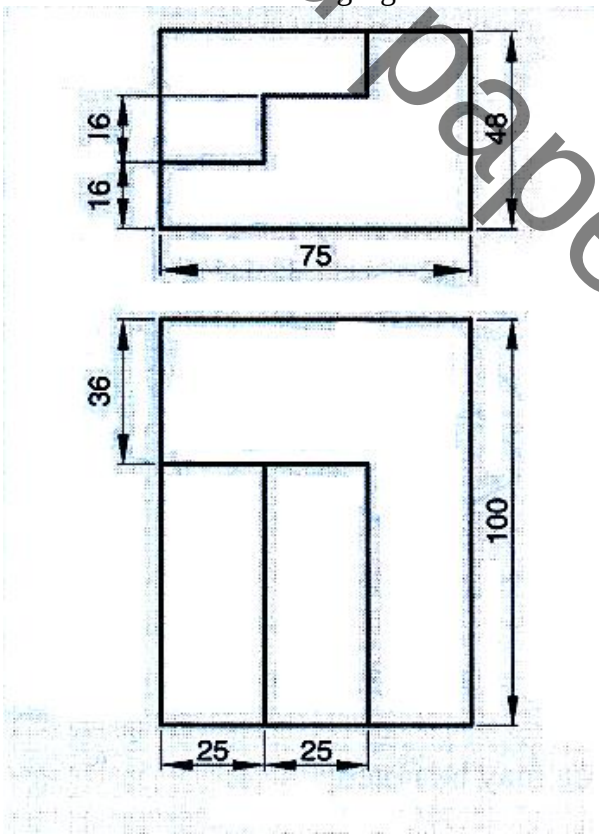


Figure 2

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